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The formed surface characteristics of SiC_f/SiC composite in the nanosecond pulsed laser ablation

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ABSTRACT

For the advantages of high-temperature resistance, corrosion resistance and ultra-high hardness, SiC_f/SiC composite is becoming a preferred material for manufacturing aero-engine parts. However, the anisotropy and heterogeneity bring great challenges to the processing technology. In this study, a nanosecond pulsed laser is applied to process SiC_f/SiC composite, where the influence of the scanning speed and laser scanning direction to the SiC fibers on the morphology of ablated grooves is investigated. The surface characteristics after ablation and the involved chemical reaction of SiC_f/SiC are explored. The results show that the increased laser scanning speed, accompanied by the decreasing spot overlap rate, leads to the less accumulation of energy on the material surface, so the ablation effect drops. In addition, for the anisotropy of the SiC_f/SiC material, the obtained surface characteristics are closely dependent on the laser scanning direction to the SiC fibers, resulting in different groove morphology. The element composition and phase analysis of the machined surface indicate that the main deposited product is SiO_2 and the carbon substance. The results can provide preliminary technical support for controlling the machining quality of ceramic matrix composites.

1. Introduction

With the rapid development of the aerospace technology, the high performance requirements of aero-engine parts, such as the hot flame tubes, heat shields, combustion chambers, and so on, are putting great pressure on the development of the materials and the manufacturing process. Ceramic Matrix Composites (CMCs) are promising materials for the high specific strength and modulus, excellent wear resistance and good corrosion resistance [1]. As a typical CMCs material, the low density of CMCs-SiC can reduce the mass by up to 30%, compared with nickel-based superalloys [2], which is now experiencing wide application as thermal structures [3,4]. Two main types of CMCs-SiC composites are carbon fiber reinforced silicon carbide composites (C_f/SiC) and silicon carbide fiber reinforced silicon carbide composites (SiC_f/SiC) according to the reinforcement phases [5,6]. During the manufacturing process of the CMCs-SiC, pyrolytic carbon layer is deposited on the surface of the internal fibers by chemical vapor deposition, and the preform is then obtained by braiding process. The CMCs-SiC composites are then fabricated by the precursor infiltration and pyrolysis (PIP) process [7,8]. Even though the added toughening SiC_f phases help to overcome the fracture and brittleness of ceramic-based materials, it results in the material anisotropy to further make the precision machining of SiC_f/SiC of great difficulty, the heterogeneity and anisotropy of which have an obvious influence on the machining quality and the obtained surface characteristics [9].

Generally, the ceramic matrix composites are mainly machined with diamond tools to achieve better surface quality and dimensional accuracy. Except for the low efficiency and short tool life, surface damage of CMCs-SiC is easy to be generated, such as the burrs, the chipping, the delamination and the tearing [10,11]. Du et al. [12] and Zhang et al. [13,14] also found that the fiber fracture, interface debonding, fiber pull-out, matrix cracking were induced during the grinding process. Compared with the mechanical machining process, the high-energy laser beam can effectively avoid the surface defects by melting or breaking the chemical bonds of materials, and a number of researches on the laser machining of CMCs-SiC have been undertaken. To explore the

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feasibility in machining ceramic matrix composites, it is important to understand the interaction mechanism between laser beam and ceramic matrix composites. Jiao et al. [15] studied the material removal mechanism of the needle punched fiber reinforced ceramic matrix composites by nanosecond pulsed laser and analyzed the influence of the laser parameters on the morphology of the ablated holes, where the holes could be obtained when the laser power exceeded 25 W, accompanied by material re-solidification and fiber fracture. Pan et al. [16] carried out single point ablation of 2D C_f/SiC composites and reported that the transition region and boundary region decreased with increasing number of pulses when the power density reached a certain value and the number of pulses was more than 50. Wu et al. [17] conducted laser processing of unidirectional C_f/SiC composites with a wavelength of 1064 nm at a repetition frequency of 20 kHz, and various surface characteristics (*Sz*, *Sq*) with different fiber end orientations were obtained. Tong et al. [18] conducted basic research on the laser ablation of C_f/SiC composites by a 1 ms pulsed laser, and they presented that the ablation rate decreased with the increase of laser ablation duration and grew with the increase of power density.

The above researches are mainly aimed at the laser processing of C_f/SiC composites. However, for the different reinforcement phases, the properties of SiC_f/SiC material change. Li et al. [19] reported the picosecond laser machining of holes in SiC_f/SiC composite to study the effects of various laser parameters and different processing methods, where the spiral feed scanning path was proposed to achieve better hole quality. Chen et al. [20] investigated the laser ablation mechanism of SiC_f/SiC composites through experiments, combined with COMSOL temperature field simulation. They reported that the ablation depth and heat affected zone increased with the growth of the laser power and pulse width, as well as the drop of the pulse frequency and scanning speed. Due to the Gaussian distribution of pulse laser energy, the energy absorbed per unit area increases at a lower velocity, resulting in the increase of heat affected zone [21]. Zhai et al. [22,23] explored the ablation mechanism of femtosecond laser in processing SiC_f/SiC composite based on the finite element simulation and experiment. With the increase of the laser power, repetition frequency and scanning times, the absorption energy of material surface increased to result in more obvious ablation effect and oxidation. Although there have been a lot of researches on the laser processing of CMCs, the material removal mechanism changes due to the different composition and microstructure, such as the fiber type, the preform dimension, the weaving mode and the porosity. Therefore, to achieve better surface quality of SiC_f/SiC, it is very important to understand the dependence of the formed surface characteristics on the laser processing parameters.

In this paper, the ablation mechanism of 2.5D SiC_f/SiC composites is firstly investigated, focusing on the influence of laser scanning speed and scanning direction, and the dependence of the morphology and the quality of the groove on the scanning direction was established. The effect of scanning speed on material removal rate was then analyzed. On this basis, the induced products in SiC_f/SiC ablation are examined by XPS, EDS and Raman spectroscopy, where the oxidation mechanism of laser-processed SiC_f/SiC composites was analyzed. This study can provide guidance for quality control in machining SiC_f/SiC composite by nanosecond pulsed laser.

2. Experiments and methods

The experimental material is 2.5D SiC_f/SiC composite which is of SiC fiber, pyrolytic carbon (PyC) interface layer and SiC matrix. The pyrolytic carbon is deposited on the surface of a single fiber by chemical vapor infiltration (CVI), and the bulk material is prepared by the precursor infiltration and pyrolysis (PIP) process [24,25]. The surface and cross-section morphology of the workpiece with the porosity of about 10% is shown in Fig. 1, and the fiber diameter is about 10 μm. The material properties obtained by the specific method are listed in Table 1.

An infrared fiber nanosecond pulsed laser (YLPN-1-1 × 120-50-M, IPG Fiber Laser Technology Co., Ltd, USA) was applied for surface ablation. A schematic illustration is shown in Fig. 2, and the specific features of the equipment are listed in Table 2. The laser is transmitted through an optical fiber and a beam expander before finally irradiating on the workpiece surface through a set of focusing lens and field lens.

The workpiece is placed on a micro-displacement platform with the accuracy of 0.02 μm, and experiment was carried out in air at varying scanning speed for its obvious influence on the overlapping ratio, and the detail experimental parameters are shown in Table 3.

A laser confocal microscope (S Neox 3D optical profiler, SENSOFAR-

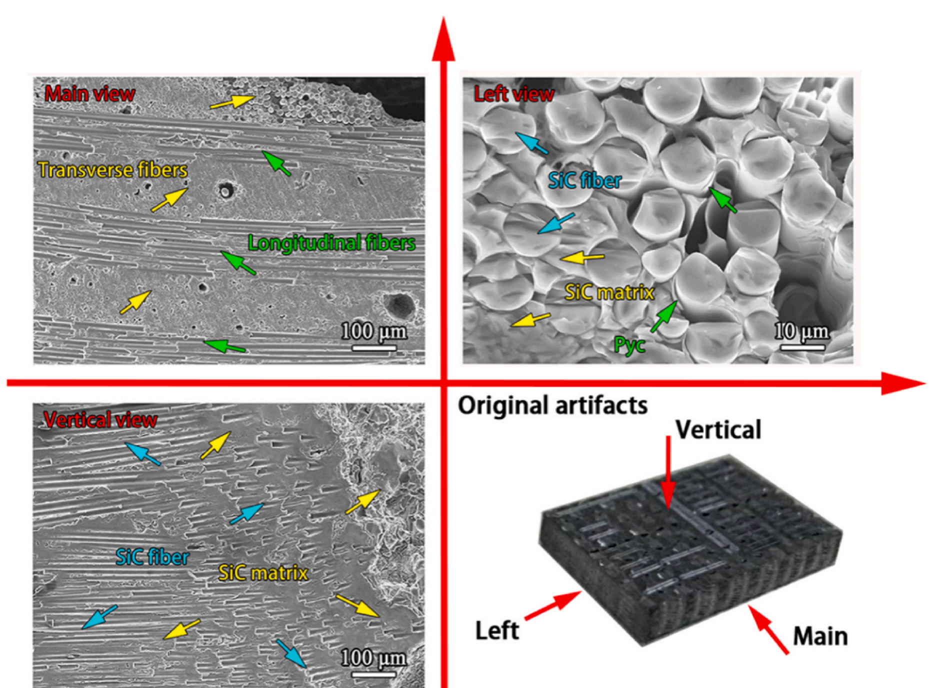


Fig. 1. SEM images of the SiC_f/SiC composite microstructure.

Table 1
The specific material parameters of SiC_f/SiC.

Parameters	Values
Workpiece size/mm	25 × 15 × 7
Density/(g/cm^3)	2.26
Porosity/%	10
Modulus of elasticity	170–230
Room temperature tensile strength/MPa	271.4
Room temperature compression strength/MPa	454.0
Room temperature bending strength/MPa	443.8
Interlaminar shear strength/MPa	63.2
Fracture toughness/MPa·m$^{1/2}$	24.1
Temperature resistance/°C	1450

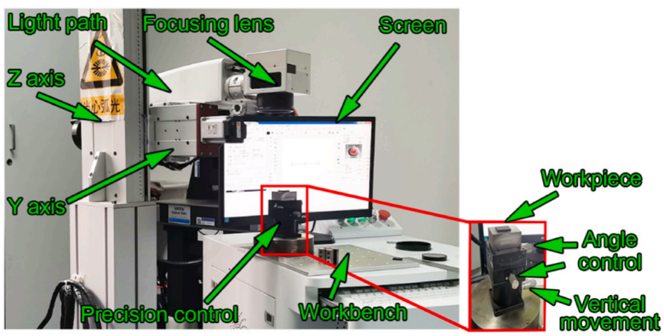


Fig. 2. Photograph of the experimental setup.

Table 2
The specific features of the laser equipment.

Parameters	Values
Nominal average output power/W	50
Nominal pulse repetition rate/kHz	50
Pulse duration/ns	120
Central emission wavelength/nm	1064
Nominal pulse energy/mJ	1.0
Laser spot size/μm	40

Table 3
The detail experimental parameters.

Parameters	Pulse frequency (kHz)	Scanning speed (mm/s)	Ablation times (N)	Power (W)
Values	20	200、400、600、800、1000	30	25

TECH SL, Spain) and a scanning electron microscope (SEM, EM-30PLUS, COXEM, Co., Ltd., Korea) were applied to examine the surface topography and morphology of the SiC_f/SiC composite before and after laser processing. The elemental composition and the phase structure of the ablated surface were then characterized by the energy dispersive spectrometer (EDS, Bruker, Co., Ltd., Germany), the X-ray photoelectron spectrometer (XPS, Kratos Analytical Co., Ltd., Japan) and R800 Raman spectrometer (HORIBA Jobin Yvon, CO., Ltd., Japan) to investigate the chemical reaction during the laser processing. In the XPS measurement, the used X-ray source is Al Kα (aluminum, unit) with the energy of 30 eV, and the C 1s binding energy is 284.8 eV for charge correction. For the Raman spectroscopy, the laser wavelength is 514 nm at the laser power of 21.5 mW with the exposure time of 20 s, and the instrument was C-calibrated before the test.

3. Results and discussion

3.1. Influence of laser scanning direction to the fibers

The surface topography and cross-sectional profile of the workpiece before and after laser ablation at the scanning speed of 200 mm/s are shown in Fig. 3. The obtained groove has errors in the symmetry caused by the parallelism error of the polished workpiece surface, the clamping error and the internal anisotropy of the SiC_f/SiC composite. The obtained groove depth and width is about 128 μm and 75 μm, respectively, as shown in Fig. 3b. The cross-sectional profile of the groove is "V" shape when the laser scanning direction is parallel to the fiber direction, and the obtained groove width and depth is about 78 μm and 128 μm, respectively, as shown in Fig. 3c. In comparison, the dimension and shape of the obtained grooves by laser scanning parallel and perpendicular to fiber direction are nearly the same.

To better understand the removal mechanism of laser ablation of SiC_f/SiC composite, the cross-section morphology of the processed groove is collected after polishing. It can be found that the inner surface of the groove is smooth, as shown Fig. 4. Fig. 4c and d shows the cross-section morphology of the grooves processed by the pulse laser parallel and perpendicular to the fiber direction, corresponding to the cross-sectional profile curves in Fig. 3b and c. In comparison, it can be found that the depth of the ablated groove results in a changing morphology at different depth and the transverse fuse is correlated with the actual fiber orientation, which is naturally dependent on the woven structure of the 2.5D SiC_f/SiC material and the laser ablation position.

Interestingly, the material removal behavior is also dependent on the fiber direction to the laser scanning. The obtained groove morphology at the laser scanning direction to the fiber directions of 0°, 30°, 60° and 90° is shown in Fig. 5. For the composite material, different ablation rates appeared due to the varying ablation threshold of the composed phases [26]. When the angle between the laser scanning direction and the fiber direction is 0°, the ablated groove edge is clean. The internal SiC matrix and Pyc interface layer are preferentially removed, and ripples appear at the inner wall of the groove along the fiber direction, as shown in Fig. 5a. Fig. 5b shows the 30° oblique appearance of the fiber end and the surface morphology, where the groove edge is not uniform. Under the varying laser scanning direction to the fiber direction, a corresponding wedge angle of the ablated fiber end face is achieved, as shown in Fig. 5c and d. The ablative traces from the surface fiber to the groove interior appear on the inner wall, which might be attributed to the preferential removal of the matrix.

3.2. Influence of scanning speed on ablation morphology

Fig. 6 shows the surface morphology of the ablated SiC_f/SiC workpiece under the different laser scanning speeds. When the scanning speed is 1000 mm/s and 800 mm/s, the depth and width of the ablation point center are approximately the same, as shown in Fig. 7a. When the scanning speed is 800 mm/s, the longitudinal fibers in the surface layer are removed, and the transverse fibers appear at the bottom. In the laser processing, the removal of SiC matrix leads to the separation of the fiber bundle, as shown in Fig. 6b. Under the scanning speed of 600 mm/s and 400 mm/s, the overlap rate increases to be 25% and 50%, which results in a more obvious ablation effect of the material, and some fibers are completely removed in the radial direction. The further drop of the scanning speed leads to a deeper ablated groove, as shown in Fig. 6c and d. As far as the scanning speed drops to be 200 mm/s, the higher overlap rate of the laser spots leads to increasing absorption of the photon energy, resulting in a more obvious ablation and material removal, as shown in Fig. 6e. Generally, the higher overlap rate, the greater energy accumulation on the material surface, leading to higher material removal volume. The lower energy accumulation results in insignificant laser ablation of the material when the laser scanning path is perpendicular to the fiber bundle at the scanning speed of 1000 mm/s, as

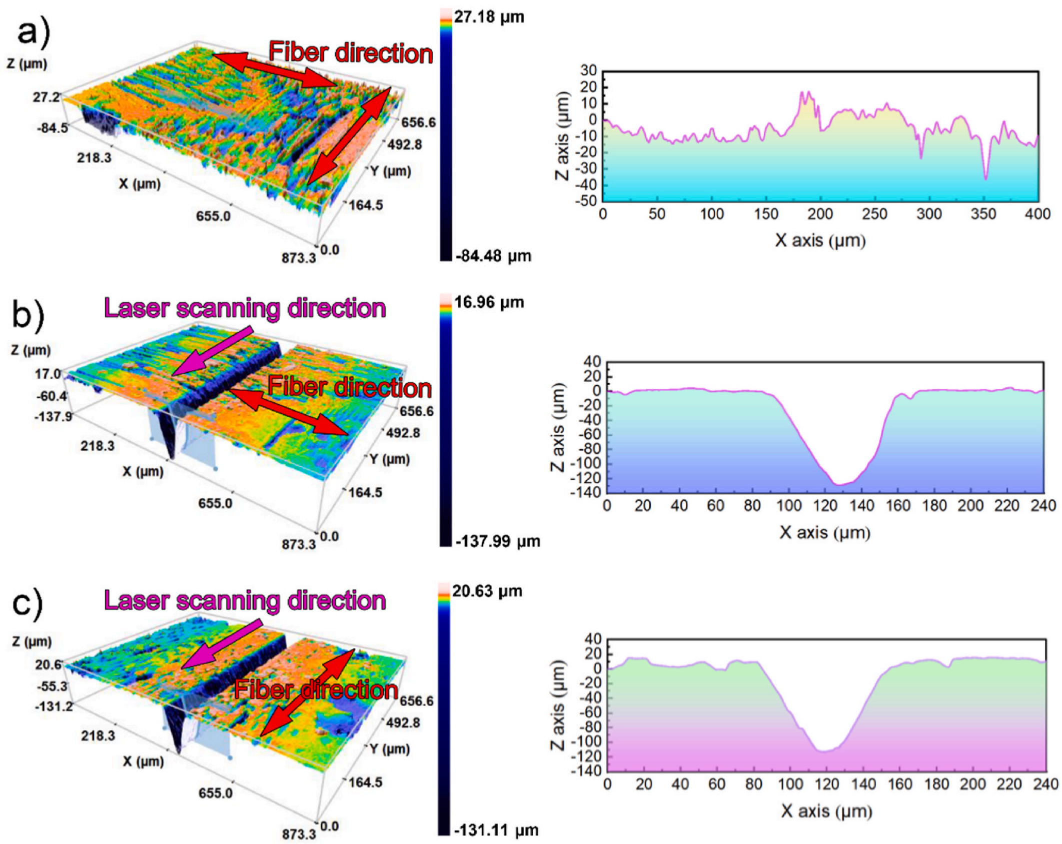


Fig. 3. Surface topography and cross-sectional profile of the SiC_f/SiC composite before and after laser ablation. (a) Original surface; (b) Laser ablation perpendicular to fiber direction; (c) Laser ablation parallel to fiber direction.

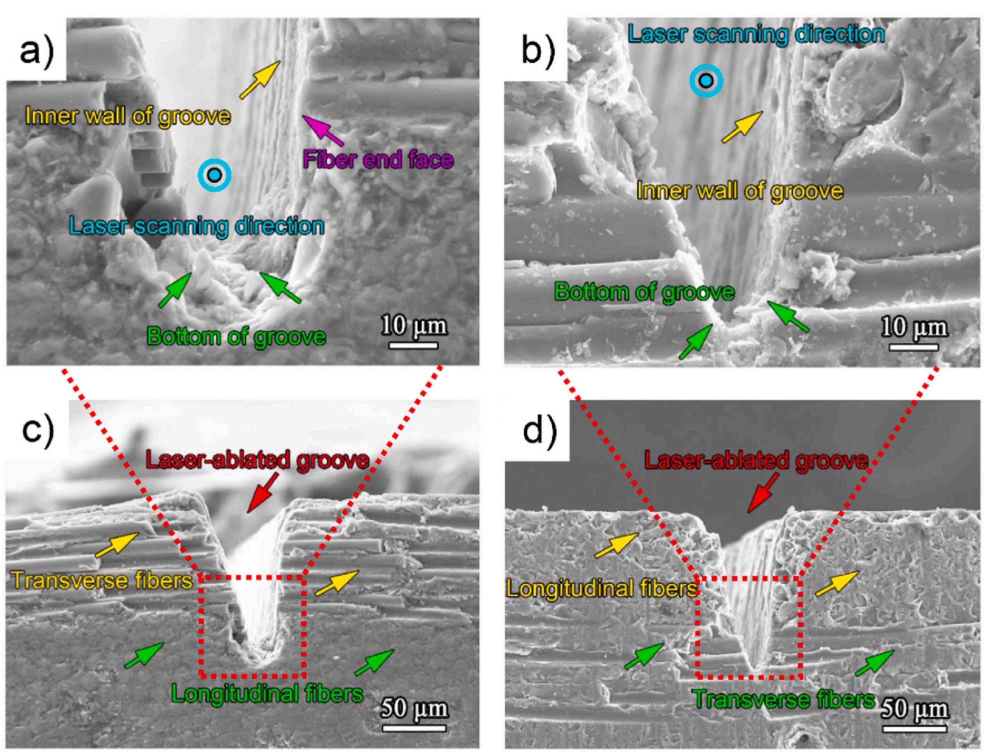


Fig. 4. The cross-section morphology of the ablated groove. (a) and (c) Laser ablation perpendicular to fiber direction; (b) and (d) Laser ablation parallel to fiber direction.

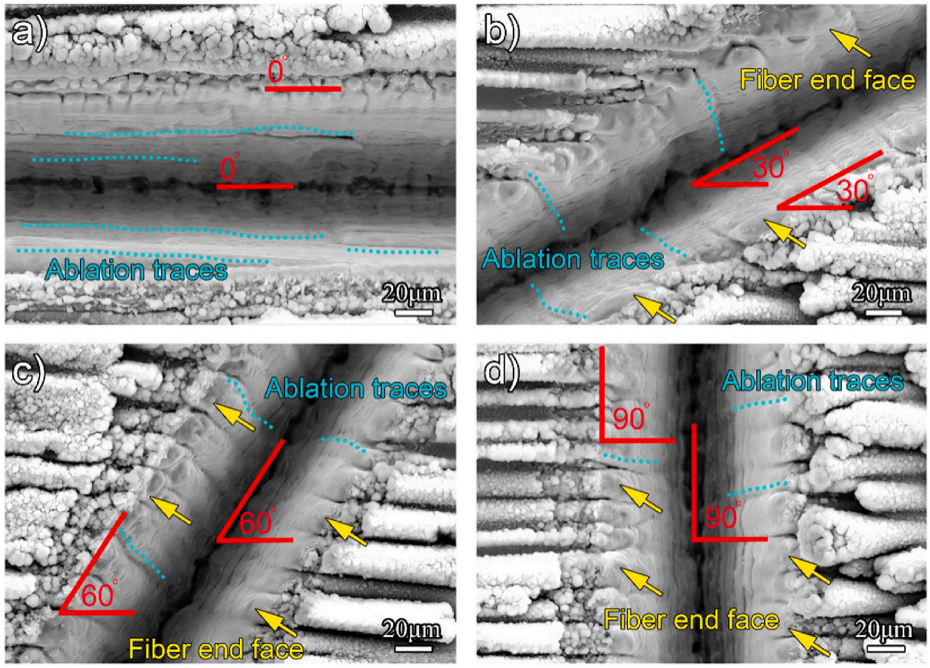


Fig. 5. SEM images of the laser ablated grooves at different laser scanning directions. (a) 0°; (b)30°; (c) 60°; (d) 90°.

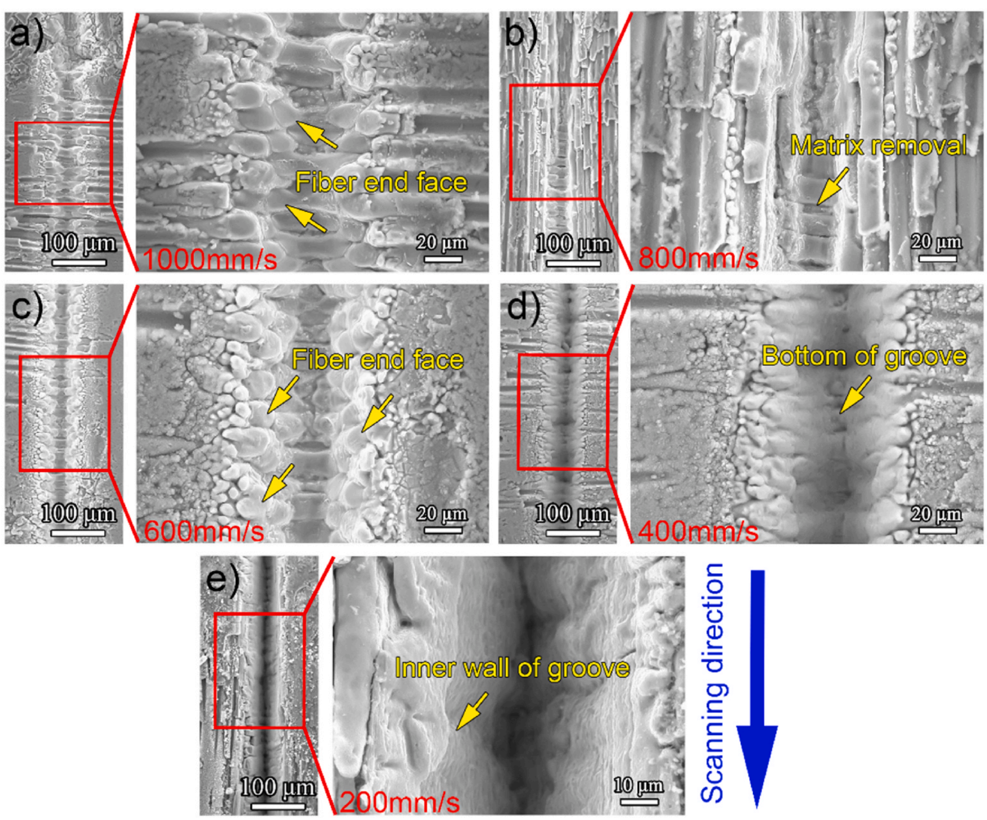


Fig. 6. SEM images of the laser ablated groove under different scanning speeds. (a) 1000 mm/s; (b) 800 mm/s; (c) 600 mm/s; (d) 400 mm/s; (d) 200 mm/s.

shown in Fig. 6a.

As the laser scanning speed increases, the ablation depth and sectional area of the groove decreases, and the material removal volume per unit length drops, as shown in Fig. 7b. The overlap rate of the laser spot changes at different scanning speed, as illustrated in Fig. 8. As the energy of a single pulse laser is constant, the accumulated energy also changes under different overlap ratio. Nanosecond pulse laser mainly removes materials by melting and vaporization, and the increase of energy accumulation leads to the growth of heat accumulation. Specifically, as the energy of a single laser pulse is 1 mJ, the growth of the overlap rate contributes to the increase of the accumulation of heat in the overlapped area, which raises the material removal efficiency.

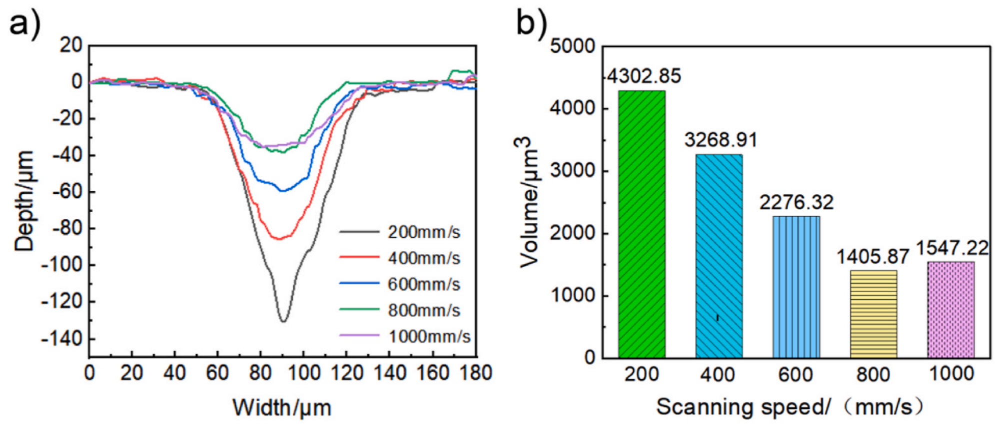


Fig. 7. The cross-sectional profiles of the ablated grooves and the material removal rates at different scanning speeds.

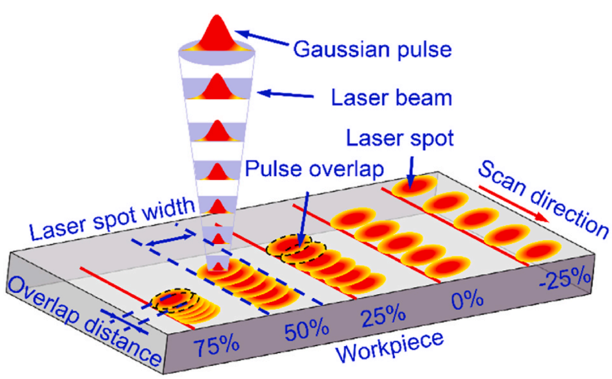


Fig. 8. Schematic illustration of the dependence of the laser pulse overlap rate on the scanning speed.

Meanwhile, for the Gaussian distribution of laser spot energy, the energy density drops rapidly along the radial direction, so the ablated degree of the fiber at the center of the spot is more serious than that at the edge.

For the applied Gaussian laser, the energy distribution varies with the overlap rate of the two spots. Therefore, the effective pulse is applied to investigate the effect of pulse overlap under different laser scanning speed and frequency [27]. The relationship between the effective pulse number and spot overlap rate and scanning times is expressed as follows [28],

$$N_e = N \cdot \frac{D}{L} = N \cdot \frac{2\omega_0}{v} \cdot f \tag{1}$$

Where N_e is the effective laser pulses, N is the laser scanning passes, f is the pulse frequency, v is the scanning speed, ω_0 is the laser spot radius, D is the spot diameter, L is the spot center distances. When the scanning frequency, pulse frequency and spot radius are constant, the effective pulse number is inversely proportional to the scanning speed, which grows with the decrease of scanning speed. Therefore, the accumulated

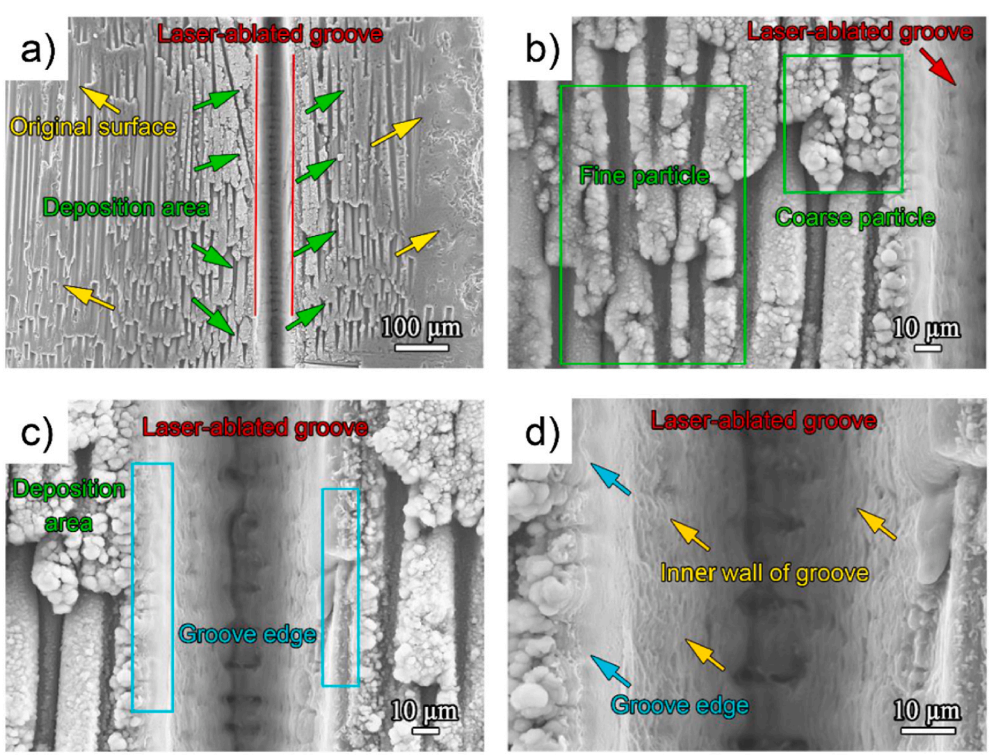


Fig. 9. SEM images of the ablated grooves parallel to fibers. (a) A whole view of the ablated area; (b) Deposited surface; (c) and (d) The groove center area.

energy per unit area increases to make the ablation more serious.

3.3. Deposition layer on ablated surface

The surface morphology and compound composition of the coagulant after laser ablation were further analyzed and studied. The surface morphology of the SiC fiber and SiC matrix after the pulsed laser processing along different fiber directions is shown in Figs. 9a and 10a, where the scanning direction of the pulsed laser is parallel and perpendicular to the fiber direction, respectively. The material is melted and vaporized as far as the concentrated energy reaches the melting point and vaporization point. The molten material splashes to the edge of the groove under the recoil pressure, and part of them re-solidified and deposited at the edge of the groove to form an oxide layer [29,30]. In the deposition area near the groove edge, the shape of the white material particles is coarse, which turns to be smaller with increasing distance, as shown in Fig. 9b. In Fig. 9c, a transition zone exists between the obtained groove and deposition area. This is attributed to the fact that the low edge energy at the edge of the Gaussian laser only results in the material melting and polishing. When the laser scanning is parallel to the fibers, the material processed with the same laser energy is relatively uniform, as shown in Fig. 9d. In addition, the inner wall of the groove is smooth, covered by the re-solidified material.

When the laser scanning direction is perpendicular to the fiber bundle, the formed surface characteristics vary, as shown in Fig. 10. Due to the different fiber directions, the coagulated coarse particles and fine particles attached to the fibers lead to different surface morphology, as shown in Figs. 9b and 10b. When the laser acts on the surface, the temperature of the material rises, and the heat propagates along the SiC fiber, the SiC matrix and the Pyc interface layer. However, both the transfer speed and the ablation threshold differ for the composed phases. Presumably, an obvious difference of the temperature field was generated. The various melting and gasification temperatures of the fiber, interfacial layer and matrix also result in different removal rates of materials under the same laser energy and the uneven morphology of processing groove edge, as shown in Fig. 10c and d. Specifically, the removal rate of the Pyc interface layer is higher than the SiC fiber, so the gullies between the fibers appeared in the transition area [31].

The ablation model was established to illustrate the surface formation mechanism, as shown in Fig. 11. The spot energy of the nanosecond pulsed laser is in a Gaussian distribution. When the spot acts on the surface of the workpiece, a large amount of laser energy is absorbed to heat the local area quickly. The thermal effect firstly occurs on the surface of the workpiece to induce oxidation. In the process of laser ablation, the material vaporizes under laser-induced high temperature, and the steam continues to absorb the laser energy. The high energy laser action on the surface of the material results in the formation of recoiling pressure in the machining area, which leads to the ejection of materials from the molten pool, leaving a crater on the surface [32]. After several hundred picoseconds, a high temperature and dense plasma formed on the surface of the material [33], where the heat accumulation of nanosecond laser still promotes the local temperature to reach the melting point of the material, forming a molten pool of a certain size. In the following process, the plasma generated on the material surface absorbs part of the laser energy through inverse bremsstrahlung radiation, thus reducing the laser ablation effect. Meanwhile, the further expansion of plasma effectively prevents the laser from reaching the material surface and affects the laser ablation effect [34].

3.4. Chemical reaction and composition change

The EDS, XPS and Raman spectroscopy were undertaken to characterize the white particles at the groove edge. Fig. 12 shows the element mass percentage of the original workpiece SiC fiber bundle by the EDS. The EDS results show that the surface adsorption of O_2 results in a little amount of O besides the detected Si and C element. As the laser ablation experiment is carried out in air, the oxygen reacts with SiC under high-temperature conditions to generate carbon-oxygen compounds and silicon-oxygen compounds. The formed gaseous oxides are mainly CO, CO_2 and SiO, and the solid SiO_2 re-solidified at the edges of the groove, as shown in Figs. 9 and 10. At the same time, SiC decomposition occurs at high temperature to produce Si and C. The Pyc layer and SiC are of

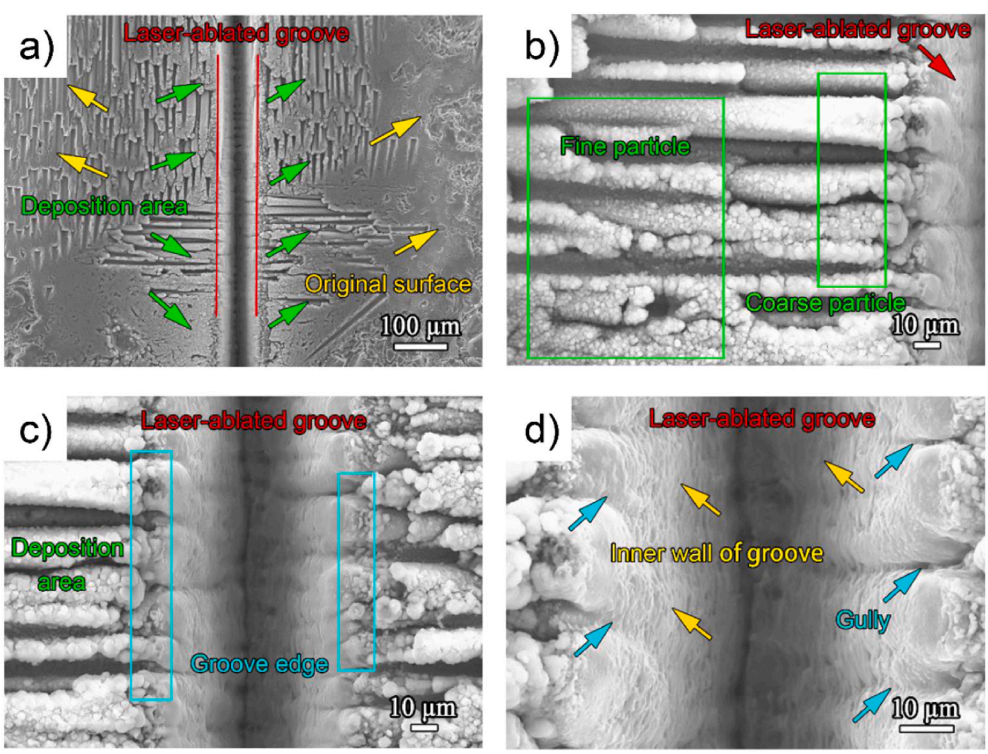


Fig. 10. SEM images of the ablated grooves perpendicular to fibers. (a) A whole view of the ablated groove; (b) Deposited surface; (c) and (d) The groove center area.

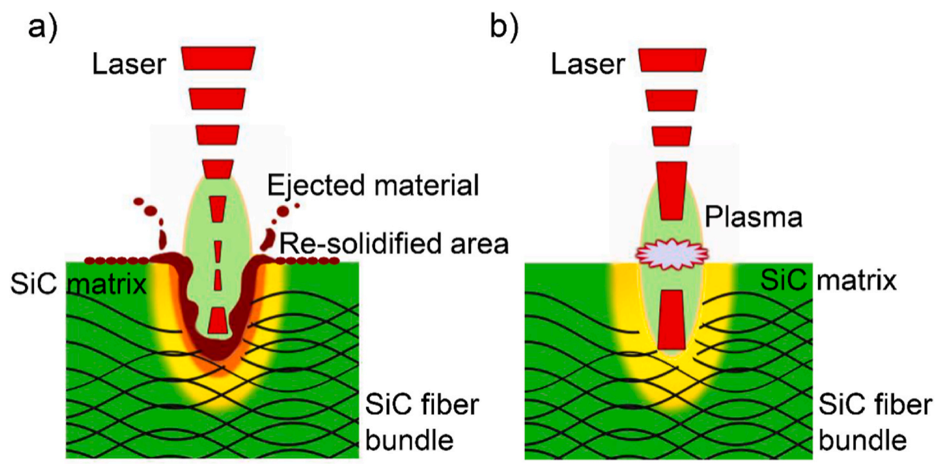
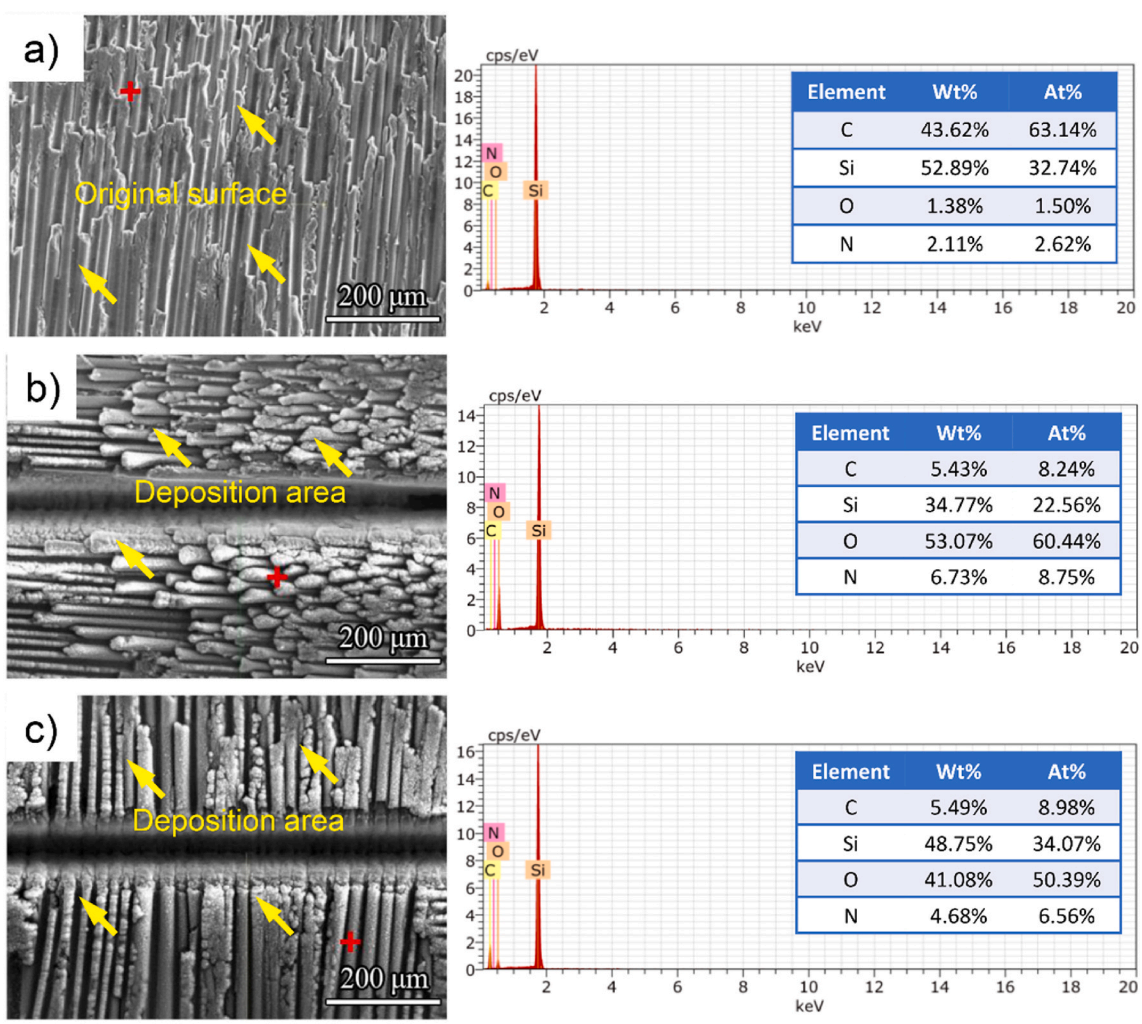


Fig. 11. The schematic diagram of the ablation process. (a) Melting removal; (b) Laser induced plasma.



Element	Wt%	At%
C	43.62%	63.14%
Si	52.89%	32.74%
O	1.38%	1.50%
N	2.11%	2.62%

Element	Wt%	At%
C	5.43%	8.24%
Si	34.77%	22.56%
O	53.07%	60.44%
N	6.73%	8.75%

Element	Wt%	At%
C	5.49%	8.98%
Si	48.75%	34.07%
O	41.08%	50.39%
N	4.68%	6.56%

Fig. 12. SEM images and EDS spectrum of (a) Original surface; (b) Fiber bundle parallel to laser scanning direction; (c) Fiber bundle perpendicular to laser scanning direction.

various oxidation reactions during laser processing under the different oxygen partial pressure. Therefore, the chemical reaction between the SiC composites and oxygen under high-temperature are proposed as follows [35],

$$SiC(s) = Si(g)+C(g) \tag{2}$$

$$C(s)+O_2(g) = CO_2(g) \tag{3}$$

$$2C(s)+O_2(g) = 2CO(g) \tag{4}$$

$$SiC(s)+2O_2(g) = SiO_2(s)+CO_2(g) \tag{5}$$

$$SiC(s)+O_2(g) = SiO(g)+CO(g) \quad (6)$$

$$2SiC(s)+3O_2(g) = 2SiO(g)+2CO_2(g) \quad (7)$$

$$2SiO_2(s)+SiC(s) = 3SiO(g)+CO(g) \quad (8)$$

Fig. 12b shows the SEM image and EDS results of the ablated grooves when the laser scanning direction is parallel to the fiber bundle direction. The mass percentage of the composed element for the resolidified layer is Si 34.77%, O 53.07%, N 6.73%, C 5.43%, respectively. By comparing the mass percentage of each element to the original material, C element on the ablated surface decreases by 38.19%, Si element decreases by 18.12%, O element increases by 51.69%, and N element has no significant change, which indicates the chemical reaction between the composites and air. Fig. 12c shows the EDS result of the white oxide at the edge of the ablated groove when the scanning direction is perpendicular to the fiber bundle. The mass percentage of the main element is Si 48.75%, O 41.08%, C 5.49%, N 4.68%, and the changing trend of the element proportion is even the same with that shown in Fig. 12b. In addition, the deposition area at the edge of the groove is mainly covered by the fine particles, which is loose, porous and of a fine snow-flake shape, so it is easy to absorb more N_2, leading to the increase of the content of N element.

Fig. 13a shows the detection positions on the original SiC_f/SiC surface, the processed SiC_f/SiC surface and the bottom of the groove by Raman spectroscopy. Fig. 13b shows the obtained spectrum of the Pyc interface layer and SiC matrix. The SiC matrix shows a strong peak value at 780 cm^{-1}, and the interface Pyc layer mainly shows D and G peaks, indicating the existence of the amorphous C. There are D and G bands of the re-solidification area on the processed material surface around 1350 cm^{-1} and 1590 cm^{-1}, indicating the presence of amorphous C. The peaks near 500 cm^{-1} for the re-solidification surface are identified to be SiO_2, and the D and G bands of amorphous C also appeared, as shown in Fig. 13c and d. At the bottom of the groove, only the D and G bands of amorphous C are present, indicating the precipitation of elemental C during the laser processing.

The composition and chemical state of the products on the machined surface were also analyzed by X-ray photoelectron spectroscopy. The three main elements of C, O, and Si are identified in the full spectrum of the measurement, as shown in Fig. 14a. The typical peak for Si_3N_4 in XPS is between 397.50 eV and 398.26 eV, and the characterized peaks of Si–N is around 150 cm^{-1} and 250 cm^{-1}. However, no compounds containing N element were found in the Raman spectra and XPS results. Therefore, only O 1s, Si 2p and C 1s were analyzed. The C 1s spectrum, Si 2p spectrum, and O 1s spectrum was analyzed, respectively, as shown in Fig. 14b, c, and d. After the nanosecond pulsed laser ablation, the C 1s spectrum showed three peak positions at 284.8 eV, 286 eV, and 288.5 eV, corresponding to the C═C, C–O, and O–C═O bonds, and the Si 2p spectrum at 103.86 eV corresponds to the Si–O bond due to the chemical reaction between the Si element with O in the air to form SiO_2 on the surface. In the O 1s spectrum, the peak positions of 533.2 eV and 535.5 eV correspond to Si–O bonds and the O═O [36]. The obtained results are in good consistence with the reported XRD analysis by Chen et al. [20].

4. Conclusion

In this study, an infrared nanosecond pulsed laser with a wavelength of 1064 nm is applied to process the SiC_f/SiC composite, and the groove morphologies after machining in different scanning directions and scanning speed are analyzed, as well as the related chemical products. The following conclusions are drawn,

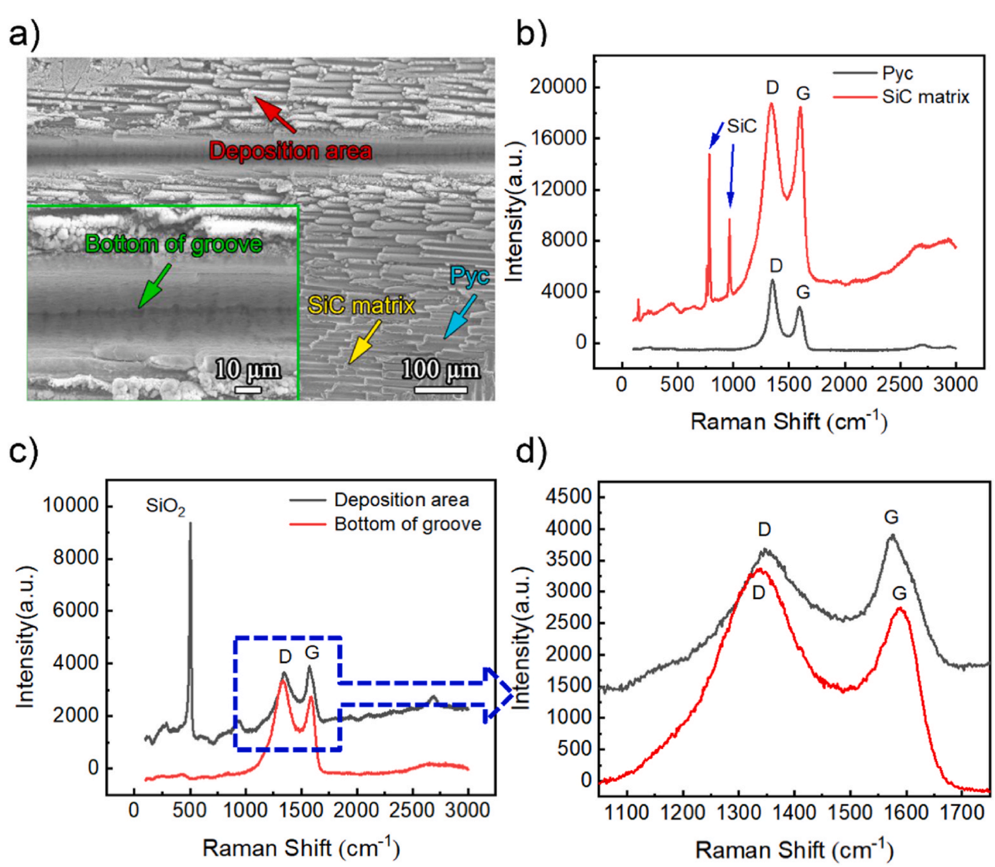


Fig. 13. Raman spectroscopy of SiC_f/SiC surface before and after the laser machining. (a) SEM image of the detection position; (b) Raman spectra of Pyc interface layer and SiC matrix before processing; (c) Raman spectra of groove bottom and deposition area after processing; (d) A local magnification in (c).

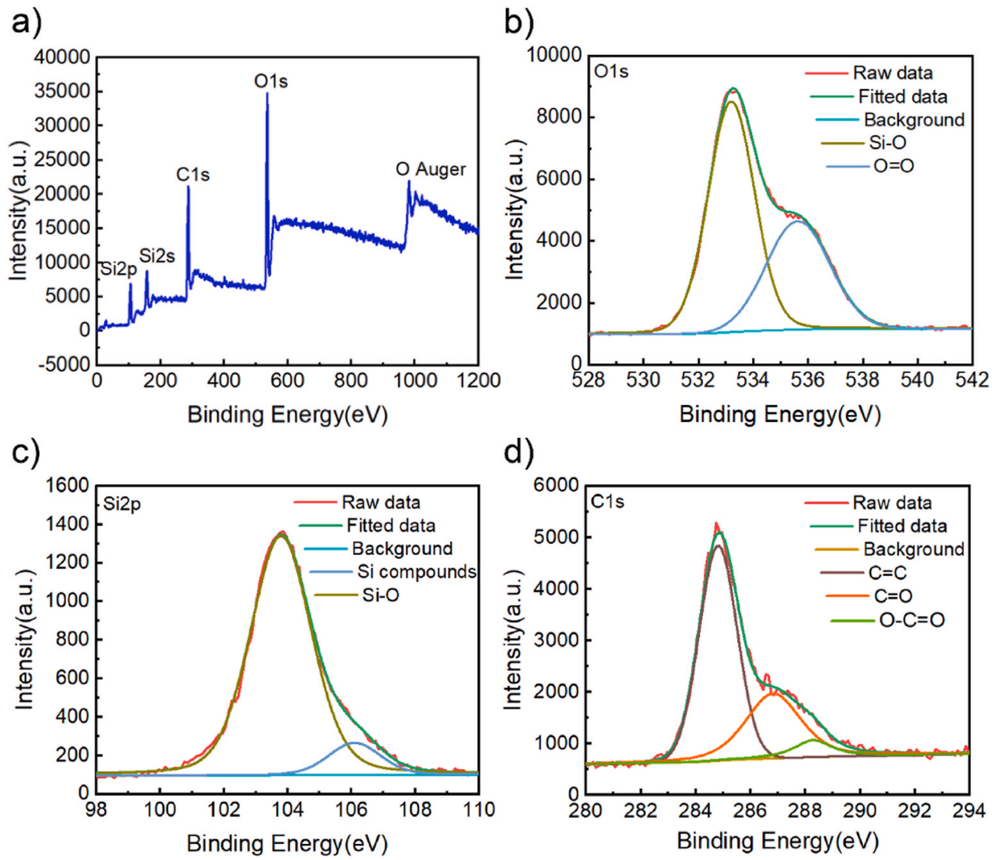


Fig. 14. XPS spectrum of the ablated surface after laser processing. (a) Full XPS spectrum; (b) O1s peak separation; (c) Si2p peak separation; (d) C1s peak separation.

(1) The overlap rate of the spot during nanosecond laser processing has a greater influence on the quality of the processed groove. With growing frequency and dropping scanning speed, accompanied by the increase of the laser spot overlap rate, the energy accumulation per unit area increases, resulting in a significant increase in the ablation depth of the groove and the change of the surface morphology.
(2) The gasification temperature of pyrolytic carbon and substrate between fibers is lower than that of SiC fiber, and the material is preferentially removed during laser ablation, where gullies appear between the fibers, resulting in poor quality of grooves edge. The morphology of groove edge and fiber section obtained by laser ablation are dependent on the relative angle between the scanning direction to the fibers, as well as the woven structure of the 2.5D SiC_f/SiC material and the laser ablation position.
(3) The decomposition of SiC and Pyc is caused by laser ablation, and graphite carbon precipitates from the inner wall of the groove. Under the action of recoiling pressure, the surface melted material splash and coagulate to form white oxidized products, where the main components are identified to be SiO_2 and carbon substance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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